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From a molecule to a useful explanation

What the new ChemEase values mean, where they come from, and how to explain them without overstating the result.

The central idea

A chemical structure becomes a graph of atoms and bonds. RDKit reads that graph and calculates descriptors. The LogD box then combines the calculated LogP with a pH, pKa and acid/base model that you enter.

Feature	Plain-language question	Type / unit
LogP	How strongly does the neutral form prefer octanol to water?	Calculated estimate / no unit
LogD	How does that preference change at a chosen pH?	Conditional estimate / no unit
TPSA	How much polar surface is represented by the molecular graph?	Calculated area / $Å^2$
HBD / HBA	Where can it donate or accept a hydrogen bond?	Two counts
Rotatable bonds	How many bonds qualify as flexible links under the model?	Count
Molar refractivity	How polarizable is the molecule estimated to be?	Calculated estimate / cm^3/mol
Formal charge	What is the sum of the atom formal charges?	Signed count / +, 0 or -
Ring count	How many rings does the toolkit perceive?	Model-dependent count
Heavy atom count	How many atoms are not hydrogen?	Count

The workflow: paste SMILES or draw a structure > validate > analyze > edit and compare > copy SMILES or download a report. SMILES is a structure notation, not a descriptor or a unique name.

How to say it

"These features turn the drawing into a molecular profile. Some outputs are counts, some are calculated models, and LogD also depends on the conditions we enter."

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LogP: the oil-water preference

Partition coefficient describes an equilibrium concentration ratio. LogP puts that ratio on a base-10 scale.

For a **neutral molecular species**, compare its concentration in octanol with its concentration in water. Octanol is used as a standard organic phase; it is not a literal cell membrane.

The equations

```
P = C_octanol / C_water (neutral species)
LogP = log10(P)    so    P = 10^(LogP)
```

LogP	P	What the number says
-1	0.1	Water concentration is 10 times the octanol concentration.
0	1	The concentrations in the two phases are equal.
1	10	Octanol concentration is 10 times the water concentration.
2	100	Octanol concentration is 100 times the water concentration.

How ChemEase gets it

The app uses the **Wildman-Crippen atom-contribution model**. Atom types are assigned from their chemical environments, and model contributions are added. This is a structure-based estimate, not a laboratory partition experiment. [R1]

```
Estimated LogP = sum of assigned atom contributions
```

Worked examples from this ChemEase build

Molecule	LogP	10^LogP = P
Ethanol	-0.0014	1.00
Benzene	1.6866	48.60
Aspirin	1.3101	20.42

Example: aspirin gives LogP 1.3101, so P = 10^1.3101 = 20.42. Its calculated neutral-form octanol concentration is about 20 times its water concentration at equilibrium. This is not a percentage of molecules or a prediction of absorption.

How to say it

"A one-unit increase in LogP means a tenfold increase in the concentration ratio. Higher is more lipophilic; it does not automatically mean better, more soluble, or safer."

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LogD: the same question, at a pH

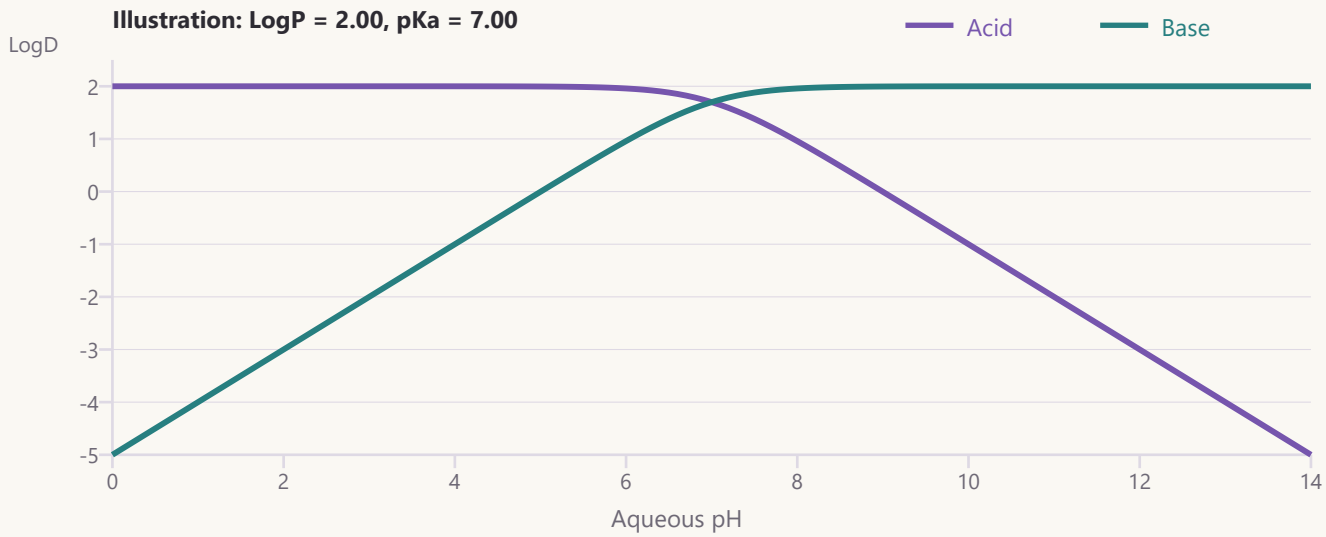
Ionization changes the mixture of neutral and charged forms. LogD describes the distribution of all forms at the stated pH. [R4]

pKa tells us where acid/base forms balance. For HA / A- or BH+ / B, pH = pKa means equal concentrations of the two forms. Under the app's model, only the neutral form partitions into octanol.

ChemEase equations: one ionizable site

$$\text{Acid: } LogD = LogP - \log_{10}(1 + 10^{pH - pKa})$$

$$\text{Base: } LogD = LogP - \log_{10}(1 + 10^{pKa - pH})$$



Read the curves: an acid becomes more ionized as pH rises, lowering LogD. A base becomes less ionized as pH rises, raising LogD. At pH = pKa, both give LogD = LogP - 0.301. The figure uses hypothetical inputs, not a measured compound.

A calculation you can explain step by step

Suppose a monoprotic acid has **LogP = 2.00**, **pKa = 4.00** and **pH = 6.00**. The ionized-to-neutral ratio is 10^(6 - 4) = 100. The neutral fraction is 1/101. Therefore D = 100/101 = 0.9901 and **LogD = -0.0043**.

In the app: enter pH, pKa and acid/base type yourself. It does not predict pKa. The estimator accepts one neutral, connected molecule; it is disabled for ions, salts and zwitterions. Multiple ionizable sites or appreciable partitioning of ions require a richer model.

How to say it

"LogP describes the neutral form. LogD adds the pH-dependent ionization balance, so we must report the pH, pKa and model with the number."

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Polarity and hydrogen bonding

TPSA, HBD and HBA describe related aspects of a molecule, but they measure different things.

TPSA: topological polar surface area

TPSA is a fragment-based area calculated from the two-dimensional molecular graph. RDKit adds contributions for recognized polar atom environments. It is reported in **square angstroms (Å²)**. It is not the total surface of the molecule, and it does not require the 3D viewer. [R2]

Aspirin's TPSA adds up: two carbonyl oxygen contributions (17.07 each), ester O (9.23) and acid OH (20.23) give **17.07 + 17.07 + 9.23 + 20.23 = 63.60 Å²**. These values apply to these oxygen environments. [R6] TPSA alone does not determine solubility or permeability. Sulfur/phosphorus options can differ across tools.

HBD and HBA: count the sites, not every N or O

HBD: an atom that can donate an attached hydrogen in a hydrogen bond. An -NH2 group normally counts as one donor site, not two donor hydrogens.
HBA: an atom with an available electron pair that can accept a hydrogen bond. Charge, aromaticity and resonance matter. [R3]

Molecule	TPSA (Å²)	HBD	HBA
Benzene	0.00	0	0
Ethanol	20.23	1	1
Acetic acid	37.30	1	1
Acetamide	43.09	1	1
Aspirin	63.60	1	3

Why the examples differ

Ethanol: the hydroxyl oxygen contributes polar surface and can donate or accept a hydrogen bond: HBD 1, HBA 1.
Acetic acid: the acid OH is a donor; the carbonyl O is an acceptor. The acid OH is not counted as an acceptor by this model.
Acetamide: the -NH2 nitrogen is a donor, but its electron pair is delocalized toward the carbonyl, so it is not an acceptor. The carbonyl O is an acceptor.

How to say it

"TPSA is an area. HBD and HBA are counts of eligible interaction sites. We look at the atom's chemical environment, not just its element symbol."

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Flexibility, rings and molecular size

These descriptors summarize how the atoms are connected. They do not require a simulation of molecular motion.

Rotatable bonds: a defined count of flexible links

ChemEase uses RDKit's strict rotatable-bond count. Eligible bonds are typically non-ring single bonds between non-terminal heavy atoms; restrictions such as amide resonance reduce the count. [R3]

Example	Rotatable bonds	Reason to explain
n-Butane, CCCC	1	Only the central C-C qualifies; terminal bonds are excluded.
Ethanol, CCO	0	The strict count is zero even though real bonds can move.
Cyclohexane, C1CCCCC1	0	Ring bonds are excluded; the ring can still change conformation.
Aspirin	2	Two bonds qualify under the toolkit's strict definition.

Ring count: perceived rings, not every closed route

A ring is a closed connection through the molecular graph. RDKit reports 1 for benzene and 2 for naphthalene. Fused and bridged systems make ring definitions less obvious; a count is not a list of every possible closed path. The separate *Independent rings* field uses graph cycle rank, E - V + connected components. For simple connected examples the counts agree; complex cases can differ. [R2]

Heavy atom count: every atom except hydrogen

Carbon, nitrogen, oxygen, halogens and other non-hydrogen elements are heavy atoms for this count. Implicit hydrogens still contribute to formula and mass, but not to the heavy atom count. [R3]

Aspirin makes the counting clear

Formula $C_9H_8O_4$ gives **9 carbon + 4 oxygen = 13 heavy atoms**. Including hydrogen gives 21 total atoms. The counts answer different questions.

How to say it

"Rotatable bonds summarize eligible flexibility, rings summarize connectivity, and heavy atoms summarize non-hydrogen size. Zero rotatable bonds does not mean zero molecular motion."

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Molar refractivity and formal charge

One is a model related to the electron cloud; the other is bookkeeping assigned to atoms in a structure.

Molar refractivity: a polarizability-related descriptor

Polarizability describes how easily an electron cloud can be distorted by an electric field. Molar refractivity is related to this response and molecular volume. ChemEase reports a **Wildman-Crippen estimate** built from atom contributions, in **cm^3/mol**. It does not measure a sample's refractive index. [R1]

Do not confuse the model with an experiment

In physical measurements, molar refraction is related to refractive index n, molar mass M and density rho by:

$R = [(n^2 - 1)/(n^2 + 2)] \times (M/rho)$

ChemEase estimates the descriptor from structure instead of asking for n and rho. [R5]

Example	Estimated molar refractivity
Ethanol	12.76 cm^3/mol
Benzene	26.44 cm^3/mol
Aspirin	44.71 cm^3/mol

Formal charge: sum the charges assigned to atoms

For one atom, formal charge = valence electrons - nonbonding electrons - half the bonding electrons. Add the atom formal charges to get the molecule's net formal charge. Formal charge is not a partial charge or a computed electrostatic surface.

Structure	Net formal charge	Interpretation
Ammonium, [NH4+]	+1	One positively charged ion.
Acetate, CC(=O)[O-]	-1	One negatively charged ion.
Glycine zwitterion	0	A +1 and a -1 site cancel in the net sum.

How to say it

"Net zero does not always mean that every atom is uncharged. That is why the simple LogD estimator rejects zwitterions even though their net charge is zero."

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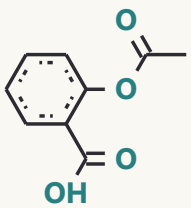
Worked examples you can reproduce

Paste the listed SMILES into ChemEase. Values below come from the deployed engine, not from a separate reference database.

Molecule	LogP	TPSA (Å²)	MR (cm³/mol)
Ethanol	-0.0014	20.23	12.76
Benzene	1.6866	0.00	26.44
Acetic acid	0.0909	37.30	13.31
Aspirin	1.3101	63.60	44.71
Caffeine	-1.0293	61.82	51.20

Molecule	HBD	HBA	Rot.	Rings	Heavy	Charge
Ethanol	1	1	0	0	3	0
Benzene	0	0	0	1	6	0
Acetic acid	1	1	0	0	4	0
Aspirin	1	3	2	1	13	0
Caffeine	0	3	0	2	14	0

MR = molar refractivity; Rot. = strict rotatable bonds. HBA uses this build's NumHBA definition. For example, caffeine gives HBA 3 here; an N+O atom count would give 6 and is a different quantity. Ethanol's near-zero LogP is shown to four decimal places to avoid a misleading rounded "-0.00".



Aspirin
`CC(=O)Oc1ccccc1C(=O)O`

Ethanol: `CCO` **Benzene:** `c1ccccc1`
Acetic acid: `CC(=O)O`
Caffeine: `Cn1c(=O)c2c(ncn2C)n(C)c1=O`

Explain aspirin in one pass

Its formula is **C9H8O4**, so it has **13 heavy atoms**. Its benzene ring gives ring count **1**. The acid OH gives HBD **1**; three eligible oxygen sites give HBA **3**. Its calculated profile is LogP **1.3101**, TPSA **63.60 Å²**, MR **44.71 cm³/mol** and **2** rotatable bonds.

A LogD demonstration: for aspirin's calculated LogP 1.3101, choose acid, pH 7.4 and an *illustrative entered pKa* of 3.5. LogD = 1.3101 - log10(1 + 10^3.9) = **-2.59**. This demonstrates the calculation; it does not establish an experimental pKa.

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A clear explanation, ready to present

A short talk track, a live demonstration, and the sources behind the definitions.

Your 60-second explanation

"ChemEase starts with a drawing or a SMILES string and turns it into an editable molecular graph. RDKit uses that graph to calculate a profile. LogP describes the neutral molecule's oil-water preference; TPSA, HBD and HBA describe polar interactions. Rotatable bonds, rings and heavy atoms summarize flexibility and connectivity. Molar refractivity relates to polarizability, while formal charge adds the charges assigned to atoms. For LogD we also enter pH, pKa and an acid/base model. These are model-based explanations, not experimental measurements or a guarantee of biological behavior."

A simple live demonstration

1. Open **SMILES**, choose **Aspirin**, validate, then open it in the editor.
2. Open **Analyze Structure** and explain the 13 heavy atoms, 1 ring and donor/acceptor counts before discussing the calculated estimates.
3. Change pH in the LogD box while keeping the entered pKa fixed. Explain the ionization assumption.
4. Copy canonical SMILES or download the properties CSV. The panel's CSV includes any valid LogD estimate and its inputs. Edit the molecule and use Undo to show that the profile follows the structure.

Answers to common questions

Are larger values always better? No. A useful profile depends on the task; no single descriptor proves activity, safety, solubility or absorption.
Why might another website disagree? Compare the exact structure, charge, tautomer, isotope, descriptor definition, toolkit version and rounding.
What happens to unsupported input? The current editor rejects stereo-specific, radical or unsupported SMILES instead of silently changing it.

Sources and reproducibility

[R1] RDKit Crippen documentation: atom-contribution LogP and MR. Open source

[R2] The RDKit Book: TPSA settings, aromaticity and ring-model context. Open source

[R3] RDKit Lipinski documentation: donors, acceptors, rotatable bonds and heavy atoms. Open source

[R4] ChEMBL compound FAQ: LogP, LogD, pKa and descriptor definitions. Open source

[R5] IUPAC Gold Book: molar refraction (M03979). Open source

[R6] RDKit TPSA source: oxygen-environment contributions used in the worked sum. Open source

All example values were regenerated with the installed ChemEase graph adapter and RDKit 2026.09.2 on 9 November 2026 (deployed code fef49b9). The LogD equations were independently evaluated for the worked examples. Values may differ from other implementations; the data in this guide intentionally match this build.